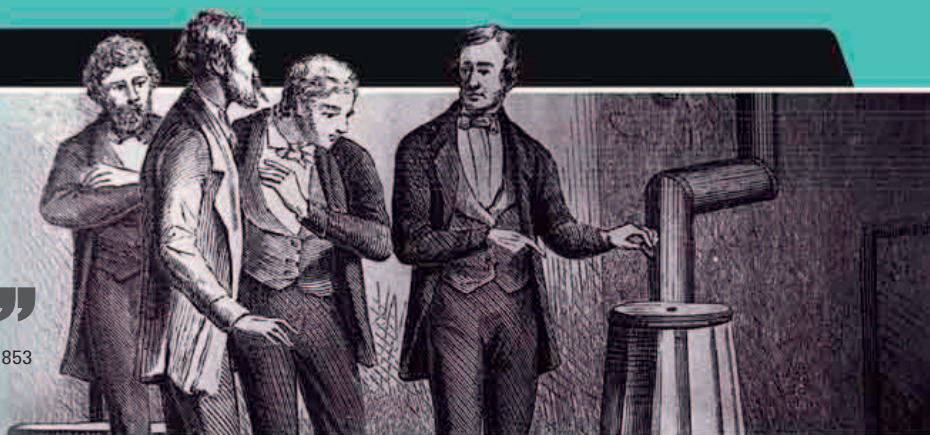


“THE INVENTOR MADE SOME EXPERIMENTS TO ASCERTAIN THE EFFECT OF HEAT... THE SPECIMEN, BEING... BROUGHT INTO CONTACT WITH A HOT STOVE, CHARRED LIKE LEATHER.”

Charles Goodyear, American inventor, in *The Applications and Uses of Vulcanized Gum-Elastic*, 1853



American inventor Charles Goodyear's experiments demonstrated that heating rubber by the right amount and adding sulfur could toughen or "vulcanize" it.

**EARLY MICROSCOPES** had been plagued by color blurring, or chromatic aberration. By 1838, achromatic microscopes—solving this problem—gave scientists a clearer view of living cells. As German physiologist **Theodor Schwann** (1810–82) studied plant and animal cells through his microscope, he realized that all living things are made of cells and cell products, and the **cell** is the **basic unit of life**.

Dutch chemist **Gerardus Mulder** (1802–80) came to an equally important conclusion about the basic material that cells are made of. After

experimenting with heating albuminous substances, such as egg white, blood, milk solids, and plant gluten with lye (a strong alkali solution), he always ended up with the same material. Mulder believed this material was composed of a **single large molecule common to all living things**. It was the Swedish chemist **Jöns Jacob Berzelius** (1779–1848) who suggested the name **protein** for this substance. It is now known there are numerous proteins and they are the basic chemicals of life.

The same year, French physicist **Claude Pouillet** (1791–1868)

made the first accurate measurements of the **solar constant** (the amount of solar heat received at Earth).

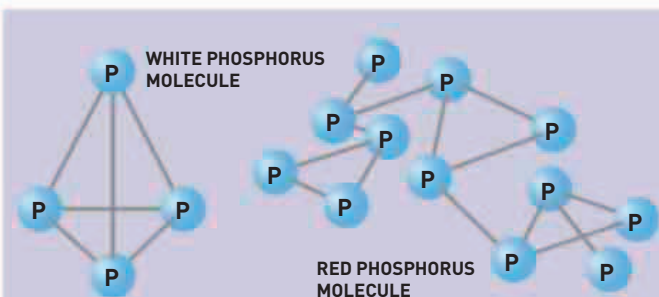
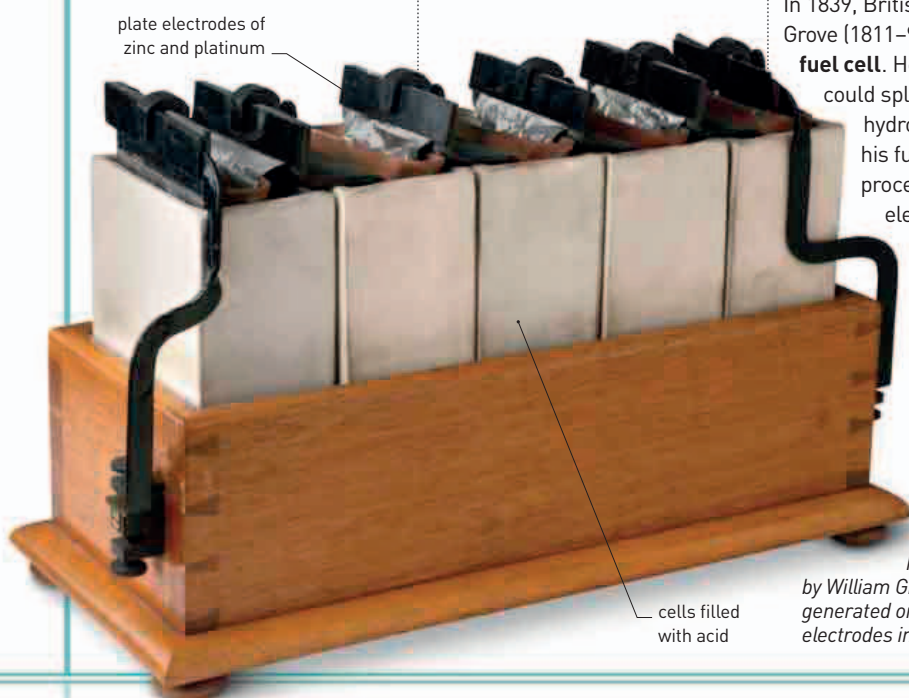
German astronomer **Friedrich Bessel** (1784–1846) made the first accurate estimate of the distance to a star using the **parallax method**, which depends on slight shifts in the star's apparent position due to the movement of the Earth.

German-Swiss chemist **Christian Friedrich Schönbein** (1799–1868) developed the idea of a fuel cell that **converts chemical energy from a fuel**, such as hydrogen, into **electricity**. In 1839, British physicist William Grove (1811–96) made the **first fuel cell**. He knew electricity

could split water into hydrogen and oxygen; his fuel cell reversed this process and made electricity by combining the two gases to produce water. At the time, though, Grove was much better known for the battery he

#### Grove cell

In this battery invented by William Grove, the charge is generated on zinc and platinum electrodes immersed in acid.



#### ALLOTROPY

Some elements come in a number of different physical forms, called allotropes. Each form is made from the same type of atom, but the atoms link up in different ways. Carbon has eight allotropes, including diamond, graphite, and fullerenes. Phosphorus has at least twelve, the most common of which are white and red solids.

invented that same year, known as a **Grove cell**.

American inventor **Charles Goodyear** (1800–60) developed a technique for vulcanizing rubber, a process that toughened it to make it suitable for use such as in tires.

In 1840, **Jacob Berzelius** suggested the word **allotrope** to describe **different forms of the same element**. Allotropes differ from each other as the result of different bonding between atoms results in

#### Solar constant

**Claude Pouillet's** 1839 estimate of solar heat radiation, made using a device called a **pyrheliometer**, was very close to today's value.

different chemical and physical properties. Also in 1840, **Christian Friedrich Schönbein** discovered an allotrope of oxygen, which he gave the name "ozone."

